Q1

In contexul unei regresii simple liniare.
Care din urmatoarele reprezinta cata variatie este/ramane nexeplicata?
(How much variation is unexplained?)

a. SSE	b. SST
c. SSY	d. SSR

Q2

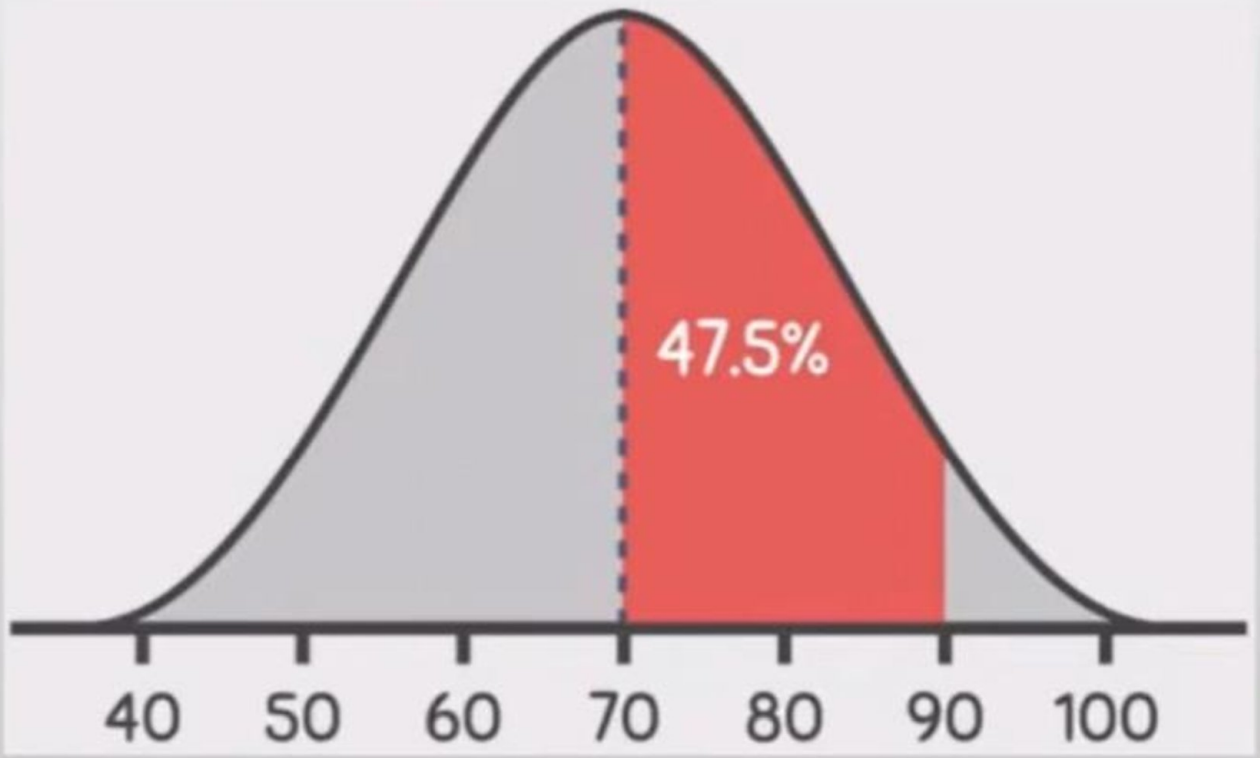
In contexul unei regresii simple liniare.
Care din urmatoarele reprezinta cata variatie este explicata de regresie?
(How much variation is accounted for by the regression?)

a. SSE	b. SST
c. SSY	**d. SSR**

ADOI (1232)

Q3

Pentru urmatoare distributie normala, cat este aria de sub grafic intre 70 si 90?



a. 47.5%	b. 40%
c. 38.5%	d. 45%

RADOI (1232)

SYSTEMS LABORATORY

Q4

EP performance crunch

Care dintre urmatoarele variante este adevarata?

X "←" Y = Y depind/e de X

a. Services ← System ← Metrics ← Workload	**b. System ← Services ← Metrics & Workload**
c. Metrics ← Workload ← System ← Services	d. Services ← System ← Metrics ← Workload

Q5

In expresia aceasta, ce reprezinta α (alpha)?

$$Probability\{c_1 \leq \mu \leq c_2\} = 1 - \alpha$$

a. Eroarea	b. Gradul de libertate
c. Probabilitatea ca $\mu \in (-\infty, c_1)(c_2, +\infty)$	d. Intervalul de incredere

Q6

In ce situatie, in formula folosita pentru calculul intervalului de incredere avem $z_{1-\frac{\alpha}{2}}$ in loc de $z_{1-\alpha}$?

$$(\overline{x} - z_{1-\frac{\alpha}{2}} \frac{s}{\sqrt{n}}, \overline{x} + z_{1-\frac{\alpha}{2}} \frac{s}{\sqrt{n}})$$

a. α este prea mare	**b. Impartim eroare de o parte si de alta a CI**
c. Avem mai putin de 30 obs in esantion	d. Toate celelalte variante

SYSTEMS LABORATORY

Q7

Pentru esantionul {-0.05, -0.10, 0.15, -0.20, -0.15, 0.10, 0.05, 0.20}, ne intereseaza un interval de incredere de 90%. In formula data, valoarea lui $\overline{x}$ este:

$$(\overline{x} - t_{[1-\frac{\alpha}{2};n-1]}\frac{s}{\sqrt{n}}, \overline{x} + t_{[1-\frac{\alpha}{2};n-1]}\frac{s}{\sqrt{n}})$$

a. 0	b. 1
c. 0.20	d. -0.20

TABLE A.4 Quantiles of the *t* Distribution

	p							
n	0.6000	0.7000	0.8000	0.9000	0.9500	0.9750	0.9950	0.999
1	0.325	0.727	1.377	3.078	6.314	12.706	63.657	636.6
2	0.289	0.617	1.061	1.886	2.920	4.303	9.925	31.59
3	0.277	0.584	0.978	1.638	2.353	3.182	5.841	12.92
4	0.271	0.569	0.941	1.533	2.132	2.776	4.604	8.610
5	0.267	0.559	0.920	1.476	2.015	2.571	4.032	6.869
6	0.265	0.553	0.906	1.440	1.943	2.447	3.707	5.959
7	0.263	0.549	0.896	1.415	1.895	2.365	3.499	5.408
8	0.262	0.546	0.889	1.397	1.860	2.306	3.355	5.041
9	0.261	0.543	0.883	1.383	1.833	2.262	3.250	4.781
10	0.260	0.542	0.879	1.372	1.812	2.228	3.169	4.587
11	0.260	0.540	0.876	1.363	1.796	2.201	3.106	4.437
12	0.259	0.539	0.873	1.356	1.782	2.179	3.055	4.318
13	0.259	0.538	0.870	1.350	1.771	2.160	3.012	4.221
14	0.258	0.537	0.868	1.345	1.761	2.145	2.977	4.140
15	0.258	0.536	0.866	1.341	1.753	2.131	2.947	4.073
16	0.258	0.535	0.865	1.337	1.746	2.120	2.921	4.015
17	0.257	0.534	0.863	1.333	1.740	2.110	2.898	3.965
18	0.257	0.534	0.862	1.330	1.734	2.101	2.878	3.922
19	0.257	0.533	0.861	1.328	1.729	2.093	2.861	3.883
20	0.257	0.533	0.860	1.325	1.725	2.086	2.845	3.850
21	0.257	0.532	0.859	1.323	1.721	2.080	2.831	3.819
22	0.256	0.532	0.858	1.321	1.717	2.074	2.819	3.792
23	0.256	0.532	0.858	1.319	1.714	2.069	2.807	3.768
24	0.256	0.531	0.857	1.318	1.711	2.064	2.797	3.745
25	0.256	0.531	0.856	1.316	1.708	2.060	2.787	3.725
26	0.256	0.531	0.856	1.315	1.706	2.056	2.779	3.707
27	0.256	0.531	0.855	1.314	1.703	2.052	2.771	3.690
28	0.256	0.530	0.855	1.313	1.701	2.048	2.763	3.674
29	0.256	0.530	0.854	1.311	1.699	2.045	2.756	3.659
30	0.256	0.530	0.854	1.310	1.697	2.042	2.750	3.646
60	0.254	0.527	0.848	1.296	1.671	2.000	2.660	3.460
90	0.254	0.526	0.846	1.291	1.662	1.987	2.632	3.402
120	0.254	0.526	0.845	1.289	1.658	1.980	2.617	3.373

SYSTEMS LABORATORY

Q8

Pentru esantionul {-0.05, -0.10, 0.15, -0.20, -0.15, 0.10, 0.05, 0.20}, ne intereseaza un interval de incredere de 90%. In formula data, valoarea lui $t_{[1-\frac{\alpha}{2};n-1]}$ este:

$$(\overline{x} - t_{[1-\frac{\alpha}{2};n-1]}\frac{s}{\sqrt{n}}, \overline{x} + t_{[1-\frac{\alpha}{2};n-1]}\frac{s}{\sqrt{n}})$$

a. 1895	b. 1860
c. 1415	d. 1397

TABLE A.4 Quantiles of the *t* Distribution

	p							
n	0.6000	0.7000	0.8000	0.9000	0.9500	0.9750	0.9950	0.999
1	0.325	0.727	1.377	3.078	6.314	12.706	63.657	636.6
2	0.289	0.617	1.061	1.886	2.920	4.303	9.925	31.59
3	0.277	0.584	0.978	1.638	2.353	3.182	5.841	12.92
4	0.271	0.569	0.941	1.533	2.132	2.776	4.604	8.610
5	0.267	0.559	0.920	1.476	2.015	2.571	4.032	6.869
6	0.265	0.553	0.906	1.440	1.943	2.447	3.707	5.959
7	0.263	0.549	0.896	1.415	1.895	2.365	3.499	5.408
8	0.262	0.546	0.889	1.397	1.860	2.306	3.355	5.041
9	0.261	0.543	0.883	1.383	1.833	2.262	3.250	4.781
10	0.260	0.542	0.879	1.372	1.812	2.228	3.169	4.587
11	0.260	0.540	0.876	1.363	1.796	2.201	3.106	4.437
12	0.259	0.539	0.873	1.356	1.782	2.179	3.055	4.318
13	0.259	0.538	0.870	1.350	1.771	2.160	3.012	4.221
14	0.258	0.537	0.868	1.345	1.761	2.145	2.977	4.140
15	0.258	0.536	0.866	1.341	1.753	2.131	2.947	4.073
16	0.258	0.535	0.865	1.337	1.746	2.120	2.921	4.015
17	0.257	0.534	0.863	1.333	1.740	2.110	2.898	3.965
18	0.257	0.534	0.862	1.330	1.734	2.101	2.878	3.922
19	0.257	0.533	0.861	1.328	1.729	2.093	2.861	3.883
20	0.257	0.533	0.860	1.325	1.725	2.086	2.845	3.850
21	0.257	0.532	0.859	1.323	1.721	2.080	2.831	3.819
22	0.256	0.532	0.858	1.321	1.717	2.074	2.819	3.792
23	0.256	0.532	0.858	1.319	1.714	2.069	2.807	3.768
24	0.256	0.531	0.857	1.318	1.711	2.064	2.797	3.745
25	0.256	0.531	0.856	1.316	1.708	2.060	2.787	3.725
26	0.256	0.531	0.856	1.315	1.706	2.056	2.779	3.707
27	0.256	0.531	0.855	1.314	1.703	2.052	2.771	3.690
28	0.256	0.530	0.855	1.313	1.701	2.048	2.763	3.674
29	0.256	0.530	0.854	1.311	1.699	2.045	2.756	3.659
30	0.256	0.530	0.854	1.310	1.697	2.042	2.750	3.646
60	0.254	0.527	0.848	1.296	1.671	2.000	2.660	3.460
90	0.254	0.526	0.846	1.291	1.662	1.987	2.632	3.402
120	0.254	0.526	0.845	1.289	1.658	1.980	2.617	3.373

Q9

Ce tip de variabila este urmatoarea?

- Orase din Romania

a. Discreta	b. Continua
c. Ordonata	d. Neordonata

Q10

Care dintre urmatoarele este adevarata?

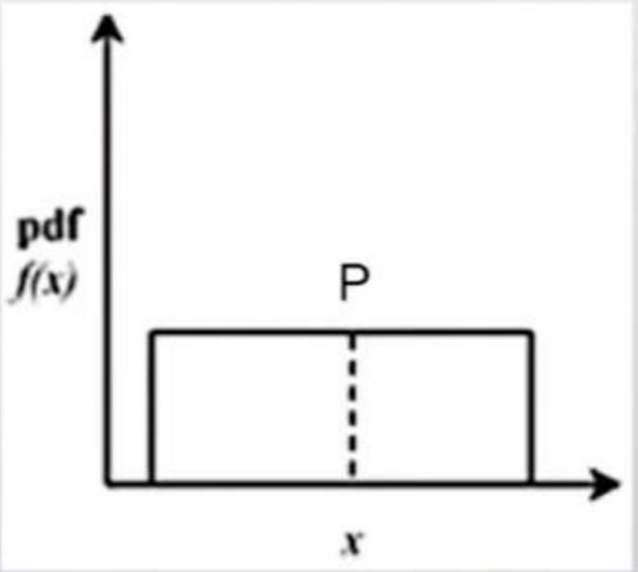
UNIX			
Parameter	**Mean**	**Std. Dev.**	**Confidence Interval**
b0	26.898	2.005	(23.2968, 30.4988)
b1	0.017	0.003	(0.0128, 0.0219)
ARGUS			
Parameter	**Mean**	**Std. Dev.**	**Confidence Interval**
b0	31.068	4.711	(22.6076, 39.5278)
b1	0.034	0.006	(0.0231, 0.0443)

a. UNIX_b0 and ARGUS_b0 are significantly different	b. UNIX_b1 > ARGUS_b1
c. UNIX_b1 and ARGUS_b1 are not significantly different	**d. UNIX_b0 and ARGUS_b0 are not significantly different**

ADOI (1232)

Q11

Ce reprezinta punctul P din graficul atasat?



a. Mean, Mode	**b. Mean, Median**
c. Doar Mode	d. Mean, Median, Mode

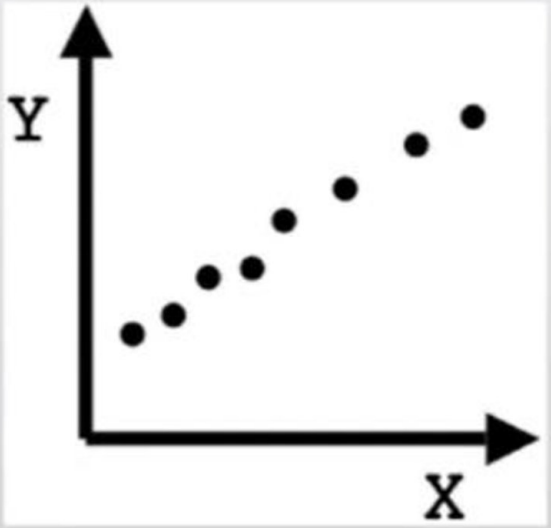
Q12

In contextul unei regresii simple liniare.
Ce inseamna $R^2 = 1$, unde R^2 este *coefficient of determination*?

a. SSR = SST	b. SSE = 0
c. Regresia / modelul este perfect	**d. Toate celelalte variante**

Q13

Ce tip de grafic este acesta?



a. Liniar	b. Discret
c. Imprastiat (scatter)	d. Continuu

ADOI (1232)

Q14

Avem un corpus de 100,000 de intrari. Fiecare intrare are ca *features* X=(nr. de ore de somn, nr. de meetinguri, nr. de minute active) si ca *labels* Y=(pulsul mediu masurat in batai pe minut).
Predictia pulsului mediu pentru o noua intrare este o problema:

a. Administrativa	b. De clasificare
c. De regresie	d. Din categoria *Unsupervised Learning*

ADOI (1232)

Q15

EP performance crunch

Sa presupunem ca am antrenat un clasificator binar care poate determina daca un fisier executabil este sau nu un virus. Modelului nostru i s-a dat executabilul "*tema_rezolvata.exe*", iar acesta NU a detectat fisierul ca fiind un virus. Din pacate, era unul.
Aceast caz este un:

a. True Positive (TP)	b. False Positive (FP)
c. True Negative (TN)	**d. False Negative (FN)**

Q16

Ce fisier trebuie editat pentru a face un swapfile persistent la reboot?

a. /etc/fstab	b. /proc/devices
c. /dev/ttyS0	d. /boot/initrd.img-$(uname -r)

- **/etc/fstab** - contine partitiile mountate la startup
- **/proc/devices** - defineste clase majore de device-uri
- **/dev/ttyS0** - device serial
- **/boot/initrd.img** - initial RAM disk, mountat inainte ca / sa fie disponibil; contine utilitare necesare procesului de bootare

Q17

EP performance crunch

De unde obtine vmstat informatiile afisate?

a. Instrumentare dinamica	b. Structuri initializate in memoria userspace prin anumite syscall-uri
c. Structuri rezidente in vDSO	**d. Fisierele de interfatare cu kernelul din /proc/**

- Ca in lab01, se vede daca facem un strace vmstat

Q18

Alegeti tipul de cache ce nu exista.

a. L1 cache	b. File cache
c. HDD cache	**d. Partition cache**

Q19

Care este principalul scop in folosirea utilitarului Wireshark?

a. Modificarea traficului de pe retea	**b. Analiza traficului de pe retea**
c. Detectarea atacurilor de tip Man-in-the-Middle	d. Implementare atacurilor de tip Denial-of-Service

Q20

Considerand formula de mai jos si valoarea deviatiei standard a erorii, putem afirma:

- Intervalul de incredere pentru 100 de predictii este mai larg decat intervalul de incredere al unei singure predictii
- Diferenta intre aceste intervale de incredere este una mica

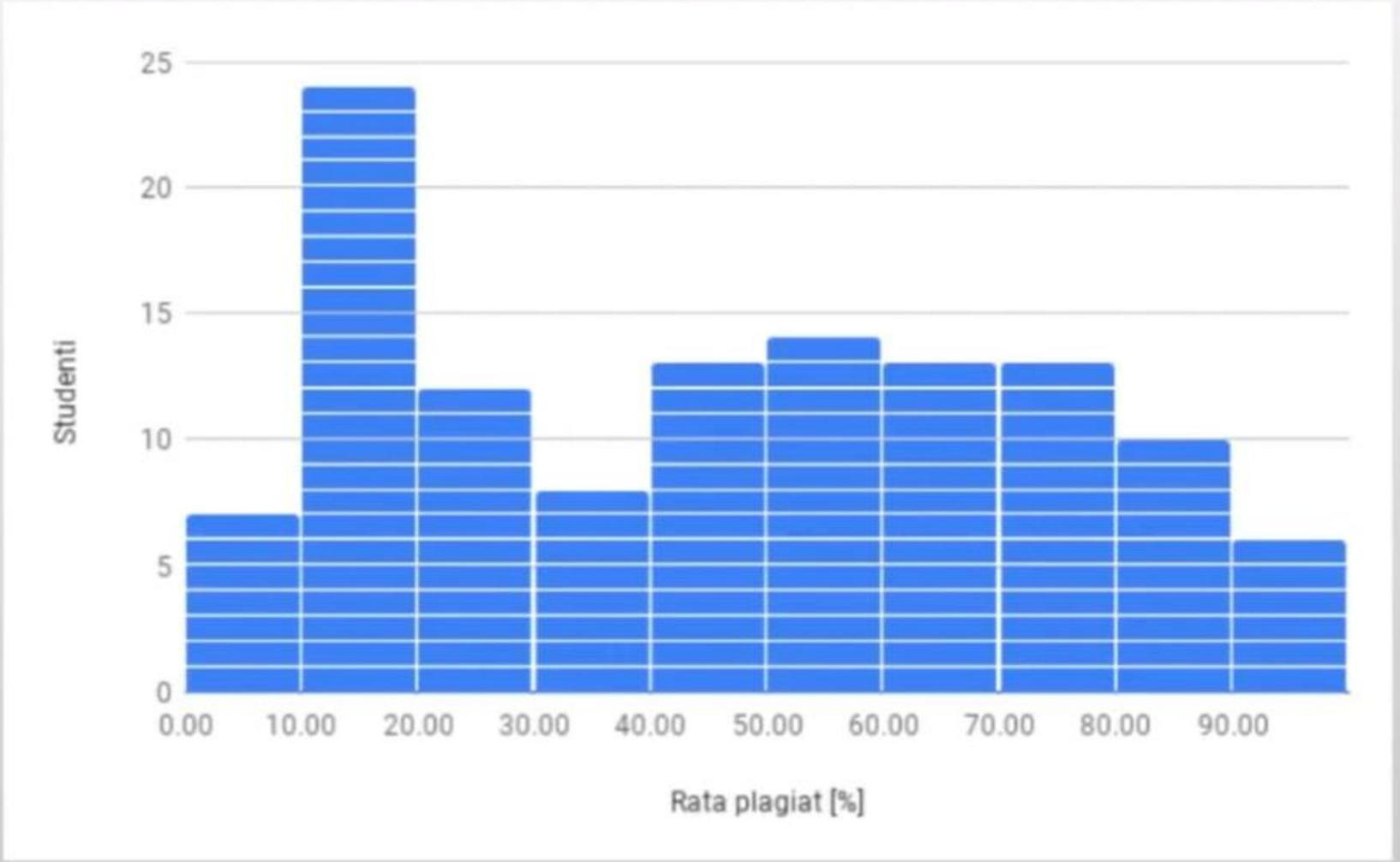
$$s_{\widehat{y}_{mp}} = s_e \left[\frac{1}{m} + \frac{1}{n} + \frac{(x_p - \overline{x})^2}{\sum x^2 - n\overline{x}^2} \right]^{\frac{1}{2}}$$

$$s_e = 1.0834$$

a. Adevarat / Adevarat	b. Adevarat / Fals
c. Fals / Adevarat	d. Fals / Fals

Q21

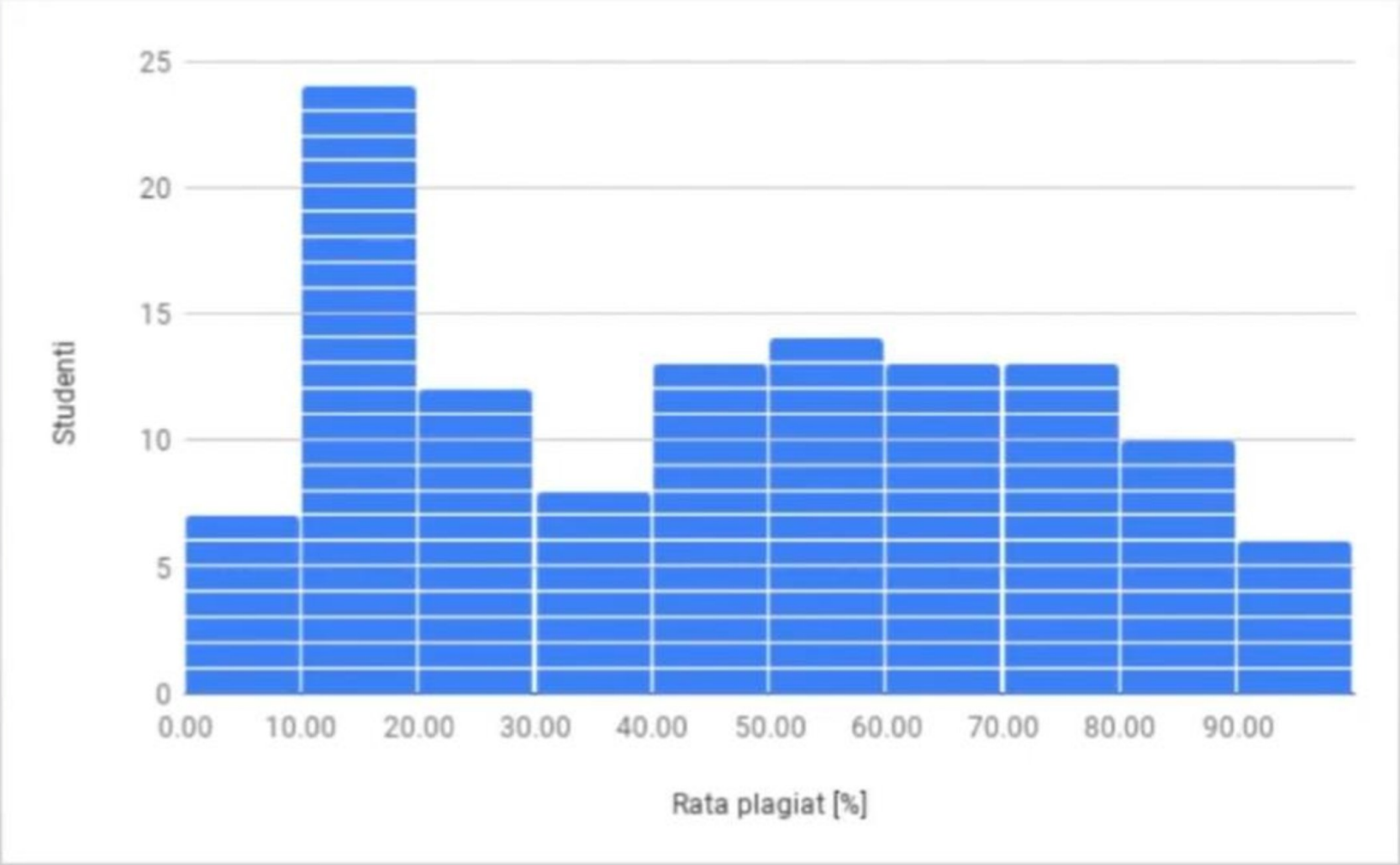
Ce tip de grafic este urmatorul?



a. Histograma	b. Stemplot
c. Scatter plot	d. Timeplot

Q22

Daca studentii cu 60% sau mai mult plagiat au luat nota zero. Un bloc reprezinta un student. Cati studenti au luat nota zero?



a. 56	b. 29
c. 42	**d. 44**